W5 – Intersections of Lines in 2-Space and 3-Space Unit 6

MCV4U

Jensen

1) Solve each linear system in two-space.

a) $3x-5y=-9,\ 4x+5y=23$

l_1 l_2

① $3x-5y=-9$

② $4x+5y=23$ +

$7x=14$

$x=2$

$3(2)-5y=-9$

$-5y=-15$

$y=3$

The solution is $(2,3)$.

b) $[x,y]=[5,4]+s[-3,1]$, $[x,y]=[2,2]+t[2,-1]$

l_1 l_2

l_1: $\begin{cases} x=5-3s \\ y=4+s \end{cases}$

l_2: $\begin{cases} x=2+2t \\ y=2-t \end{cases}$

$5-3s=2+2t$

① $3=2t+3s$

$4+s=2-t$

② $2=-t-s$

① $3=2t+3s$

2×② $4=-2t-2s$ +

$7=s$

$3=2t+3(7)$

$-18=2t$

$t=-9$

l_1: $x=5-3(7)=-16$

$y=4+7=11$

$(-16,11)$

check

l_2: $x=2+2(-9)=-16$

$y=2-(-9)=11$

2) Determine if the parallel lines in each pair are distinct or coincident.

a) l_1 $[x,y,z]=[5,-2,-8]+s[-3,2,5]$

l_2 $[x,y,z]=[-4,0,2]+t[-3,2,5]$

Check if point $(-4,0,2)$ is on l_1:

l_1: $\begin{cases} x=5-3s \\ y=-2+2s \\ z=-8+5s \end{cases}$

$-4=5-3s$ $0=-2+2s$ $2=-8+5s$

$s=3$ $s=1$ $s=2$

∴ not on l_1

∴ parallel and distinct

b) l_1 $[x,y,z]=[-3,0,-6]+s[-3,0,-6]$

l_2 $[x,y,z]=[9,1,18]+t[9,0,18]$

Check if $(-3,0,-6)$ is on l_2

l_2: $\begin{cases} x=9+9t \\ y=1 \\ z=18+18t \end{cases}$

$-3=9+9t$ $0=1$ $-6=18+18t$

$t=-\frac{4}{3}$ Not possible $t=-\frac{4}{3}$

∴ not on l_2

∴ parallel and distinct

3) Triangle ABC is formed from the intersections of the three lines represented by these equations. Find the length of each side of ΔABC.

$l_1: [x,y] = [1,-3] + t[0,1]$ $l_2: [x,y] = [2,4] + s[-1,6]$ $l_3: [x,y] = [2,3] + r[1,7]$

$l_1: \begin{cases} x = 1 \\ y = -3+t \end{cases}$

$l_2: \begin{cases} x = 2-s \\ y = 4+6s \end{cases}$

$l_3: \begin{cases} x = 2+r \\ y = 3+7r \end{cases}$

$\vec{P_1P_2} = \left[\frac{14}{13}, \frac{-84}{13}\right]$

$\vec{P_1P_3} = [0,-14]$

$\vec{P_2P_3} = \left[\frac{-14}{13}, \frac{-98}{13}\right]$

l_1 & l_2

$1 = 2-s$

$s = 1$

$x = 2-1 = 1$

$y = 4+6(1) = 10$

$P_1\ (1,10)$

l_2 & l_3

$2-s = 2+r$

① $0 = s+r$

$4+6s = 3+7r$

② $1 = -6s+7r$

6×① $0 = 6s+6r$

$1 = -6s+7r$ +

$1 = 13r$

$r = \frac{1}{13}$

$x = 2+\frac{1}{13} = \frac{27}{13}$

$y = 3+7\left(\frac{1}{13}\right) = \frac{46}{13}$

$P_2\ \left(\frac{27}{13}, \frac{46}{13}\right)$

l_1 & l_3

$1 = 2+r$

$r = -1$

$x = 2+(-1) = 1$

$y = 3+7(-1) = -4$

$P_3\ (1,-4)$

$|\vec{P_1P_2}| = \sqrt{\frac{7252}{169}} = \frac{14\sqrt{37}}{13} \simeq 6.55$

$|\vec{P_1P_3}| = 14$

$|\vec{P_2P_3}| = \sqrt{\frac{9800}{169}} = \frac{70\sqrt{2}}{13} \simeq 7.61$

4) Parallelogram ABCD has vertices A(–1, –4), B(1, –3), C(6, –6), and D(4, –7). Find the vector equations of its agonals and the point of intersection of the diagonals.

$\vec{AC} = [7,-2] = \vec{m}$

$\vec{BD} = [3,-4] = \vec{n}$

$l_1: [x,y] = [-1,-4] + t[7,-2]$

$l_2: [x,y] = [1,-3] + s[3,-4]$

$l_1 \begin{cases} x = -1+7t \\ y = -4-2t \end{cases}$

$l_2 \begin{cases} x = 1+3s \\ y = -3-4s \end{cases}$

$-1+7t = 1+3s$

① $-2 = -7t+3s$

$-4-2t = -3-4s$

② $-1 = 2t-4s$

4×① $-8 = -28t+12s$

3×② $-3 = 6t-12s$ +

$-11 = -22t$

$t = \frac{1}{2}$

$x = -1+7\left(\frac{1}{2}\right) = \frac{5}{2}$

$y = -4-2\left(\frac{1}{2}\right) = -5$

$\left(2.5, -5\right)$

5) Determine if the lines in each pair intersect. If so, find the co-ordinates of the point of intersection.

a) $l_1[x,y,z] = [9,-1,1] + s[-3,4,1]$
$l_2[x,y,z] = [-3,11,5] + t[-3,4,1]$

$l_1 \begin{cases} x = 9-3s \\ y = -1+4s \\ z = 1+s \end{cases}$ $\quad l_2 \begin{cases} x = -3-3t \\ y = 11+4t \\ z = 5+t \end{cases}$

$9-3s = -3-3t$

① $12 = 3s-3t$

$-1+4s = 11+4t$

② $-12 = -4s+4t$

$1+s = 5+t$

③ $-4 = -s+t$

① $12 = 3s-3t$
$3\times$ ③ $-12 = -3s+3t$ +
$0 = 0$

① $-12 = -4s+4t$
$4\times$ ③ $-16 = -4s+4t$ -
$4 = 0$

∴ No solutions
Parallel and distinct

b) $l_1[x,y,z] = [1,4,5] + s[3,0,-2]$
$l_2[x,y,z] = [9,4,-3] + t[3,0,-2]$

$[3,0,-2] = [3,0,-2]$; ∴ parallel

Check if $(1,4,5)$ is on l_2

$l_2 \begin{cases} x = 9+3t \\ y = 4 \\ z = -3-2t \end{cases}$

$1 = 9+3t$
$t = -\frac{8}{3}$

$4 = 4$

$5 = -3-2t$
$t = -4$

∴ not on line
∴ parallel & distinct

No solutions

c) $l_1[x,y,z] = [1,0,-3] + t[3,5,4]$
$l_2[x,y,z] = [0,-9,-1] + s[-1,2,-3]$

$l_1 \begin{cases} x = 1+3t \\ y = 5t \\ z = -3+4t \end{cases}$ $\quad l_2 \begin{cases} x = -s \\ y = -9+2s \\ z = -1-3s \end{cases}$

$1+3t = -s$

① $3t+s = -1$

$5t = -9+2s$

② $5t-2s = -9$

$-3+4t = -1-3s$

③ $4t+3s = 2$

$2\times$ ① $6t+2s = -2$
② $5t-2s = -9$ +
$11t = -11$
$t = -1$

$3(-1)+s = -1$
$s = 2$

check in ③

LS $= 4(-1)+3(2)$
$= 2$

RS $= 2$

LS = RS

POI
$x = 1+3(-1) = -2$
$y = 5(-1) = -5$
$z = -3+4(-1) = -7$

$(-2,-5,-7)$

d) ~~$[x,y,z] = [6,-4,3] + t[-2,-1,1]$~~ l_1
~~$[x,y,z] = [4,-1,2] + s[2,1,-1]$~~ l_2

$[x,y,z] = [4,7,-1] + s[-2,1,2]$
$[x,y,z] = [1,3,-1] + t[4,-1,2]$

$l_1 \begin{cases} x = 4-2s \\ y = 7+s \\ z = -1+2s \end{cases}$

$l_2 \begin{cases} x = 1+4t \\ y = 3-t \\ z = -1+2t \end{cases}$

$4-2s = 1+4t$

① $3 = 4t+2s$

$7+s = 3-t$

② $4 = -t-s$

$-1+2s = -1+2t$

③ $0 = 2t-2s$

① $3 = 4t+2s$

2×② $8 = -2t-2s+$

$11 = 2t$

$t = \frac{11}{2}$

$4 = -\frac{11}{2} - s$

$s = \frac{-19}{2}$

Check in ③

LS $= 0$ RS $= 2\left(\frac{11}{2}\right) - 2\left(\frac{-19}{2}\right) = 30$

∴ No solution
Skew lines

e) $[x,y,z] = [-2,0,-3] + t[5,1,3]$
$[x,y,z] = [5,8,-6] + s[-1,2,-3]$

$l_1 \begin{cases} x = -2+5t \\ y = t \\ z = -3+3t \end{cases}$

$l_2 \begin{cases} x = 5-s \\ y = 8+2s \\ z = -6-3s \end{cases}$

$-2+5t = 5-s$

① $5t+s = 7$

$t = 8+2s$

② $t-2s = 8$

$-3+3t = -6-3s$

③ $3t+3s = -3$

2×① $10t+2s = 14$

② $t-2s = 8$

$11t = 22$

$t = 2$

$5(2)+s = 7$

$s = -3$

check in l_3

$3(2)+3(-3) = -3$

$-3 = -3$

∴ they intersect

POI

$x = -2+5(2) = 8$

$y = 2$

$z = -3+3(2) = 3$

$(8,2,3)$

6) Determine the distance between the skew lines in each pair.

a) $\ell_1: [x,y,z] = [4,-3,2] + s[2,7,-1]$
$\ell_2: [x,y,z] = [-2,5,4] + t[-4,0,3]$

$\vec{P_1P_2} = [-6,8,2]$

$\vec{n} = \vec{m_1} \times \vec{m_2} = [21,-2,28]$

$\begin{matrix} 7 & 0 \\ -1 & 3 \\ 2 & -4 \\ 7 & 0 \end{matrix}$

$d = \left| \frac{\vec{P_1P_2} \cdot \vec{n}}{|\vec{n}|} \right|$

$= \left| \frac{-6(21)+8(-2)+2(28)}{\sqrt{(21)^2+(-2)^2+(28)^2}} \right|$

$= \left| \frac{-86}{\sqrt{1229}} \right|$

$\simeq 2.45$

b) $\ell_1: [x, y, z] = [-1,6,1] + s[-2,4,3]$
$\ell_2: [x, y, z] = [5,1,9] + t[3,-2,4]$

$\vec{P_1P_2} = [6, -5, 8]$

$\vec{n} = \vec{m_1} \times \vec{m_2} = [22, 17, -8]$

$4 \quad -2$
$3 \quad 4$
$-2 \quad 3$
$4 \quad -2$

$d = \left| \frac{\vec{P_1P_2} \cdot \vec{n}}{|\vec{n}|} \right|$

$= \left| \frac{-17}{\sqrt{837}} \right|$

$\simeq$

ANSWER KEY:

1. a) (2, 3) **b)** (−16, 11) **2. a)** Yes **b)** No **3.** $2\sqrt{37}$; 2, $10\sqrt{2}$

4. $\overline{AC} = [-1, -4] + s[7, -2]$, $\overline{BD} = [1, -3] + t[3, -4]$; $\left(\frac{5}{2}, -5\right)$

5. a) Infinitely many solutions **b)** No **c)** (−2, −5, −7) **d)** No **e)** (8, 2, 3)

6. a) 2.45 **b)** 5.15